Online Sales & Profitability

Dashboard Title

- **AdventureWorks Online Sales & Profitability Dashboard:** *Taking a deep dive towards AdventureWorks' online sales and revenue performance.*

Business Goal

- To provide executives and managers a comprehensive, real-time view of online sales performance across regions and product categories, enabling data-driven decisions about inventory management, marketing strategies, regional expansion, and resource allocation while identifying trends and profitability.

Critical Success Factor

- **Data Accuracy & Trust -** All metrics must match official financial reports and SQL data source to ensure absolute reliability. This builds stakeholder confidence and presents decisions based on flawed data.

- **Actionability -** Every visual must drive specific, tangible business decisions rather than just displaying interesting values or trivia. This ensures the dashboard actively guides inventory, marketing, and resources allocation strategies.

- **Performance & Speed -** The dashboard must load and respond to all filter interactions under 5 - 10 seconds. This guarantees a smooth, frustration-free user experience that busy executives will want to use.

- **Visual Clarity -** An executive must be able to understand the overall business health. This is achieved through clean layouts, clear hierarchies, and intuitive formatting.

- **Interactivity -** Users must be able to seamlessly drill down from high-level KPIs into granular, detailed data points. This empowers stakeholders to investigate behind the numbers without friction or difficulty.

Primary Users

- **CEO/Executives Teams:** High-level performance overview, strategic decisions.
- **Regional Managers:** Compare regional performance, allowance resources.
- **Sales Director:** Identify trends, adjust marketing strategies.
- **Inventory Managers:** Forecast demand based on sales velocity.

Data Source

- **Primary Data Source:** SQL Server - AdventureWorks Database 2025
- **Table Used:**
    - **Fact Tables:** FactInternetSales
    - **Dimensions Tables:**
        - DimDate
        - DimSalesTerritory
        - DimProduct
        - DimProductCategory
        - DimProductSubcategory
- **Views Created:** view_MastersKPIs

Filters or Slicers

- **Global Filters:**
  - **Year:** 2010-2014
  - **Month:** January-December
  - **Region:** Australia, Southwest, Northwest, etc.
  - **Country:** United States, Australia, United Kingdom
- **Visual-Level Filters:**
  - Top 3 Countries ("Top 3 Countries by Revenue")
  - Date range on trend line ("Online Sales Trend Over Time")

Lagging KPIs

- **Total Online Revenue -** This measures overall financial performance by tracking the total online sales amount generated. It serves as the primary baseline for evaluating the company's market reach and top-line growth.

- **Total Cost of Goods Sold** - This represents the historical total cost already incurred to produce or sell the products. Tracking this helps identify production inefficiencies and manage supply chain expenses.

- **Total Online Profit -** Calculated as total online revenue minus total products cost, this reflects the company's actual net earnings. It is the ultimate indicator of financial viability after covering direct expenses.

- **Total Units Sold -** This tracks the historical physical quantity of products already sold, independent of their dollar value. It is crucial for inventory management, demand forecasting and supply chain planning.

- **Profit Margin% -** This measures profitability efficiency by calculating the percentage of revenue kept as profit. A higher margin indicates better cost control and pricing power within specific categories or regions.

Leading KPIs

- **Daily Sales Run Rate -** This measures your current daily speed to project short-term financial outcomes. It acts as an early warning system to adjust tactics before the month or quarter ends.

- **Average Products Per Order -** This calculates the average number of physical items customers buy in a single transaction. It directly measures the effectiveness of cross-selling strategies and product bundling.

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- **Order per Customer -** This tracks the average number of orders placed per active customer to gauge brand loyalty. A rising trend indicates strong customer retention and successful repeat-purchase initiatives.

KPI Computations (Online Sales & Profitability Dashboard)

Lagging KPIs:

- **Total Online Profit** = Total Online Revenue - Total Cost

- **Profit Margin%** = [(Total Online Revenues - Total Cost) / Total Online Revenue] * 100

Leading KPIs:

- **Average Order Value** = Total Online Revenue / Total Number of Online Orders
- **Average Products Per Order** = Total Units Sold / Total Number of Online Orders
- **Regional Velocity** = Total Online Revenue / Total Number of Regions
- **Daily Sales Run Rate** = Total Online Revenue / 30 (Average days in a month)
- **Order per Customer** = Total Number of Online Orders / Total Number of Unique Online Customers

DAX Measures Reference

Total Online Revenue (TOR):

- SUM(view_MasterKPIs[Total_Online_Revenue])

Total Cost of Good Sold (TCOGS) :

- SUM(view_MasterKPIs[Total_Cost])

Total Online Profit (TOP):

- SUM(view_MasterKPIs[Total_Online_Revenue]) - SUM(view_MasterKPIs[Total_Cost])

Regional Velocity (RV):

- DIVIDE(SUM(view_MasterKPIs[Total_Online_Revenue]), DISTINCTCOUNT(view_MasterKPIs[Region]), 0)

Profit Margin% (PM%):

- PM% = DIVIDE([TOP], [TOR], 0)

Order per Customer (OPC):

- DIVIDE(SUM(view_MasterKPIs[Total_Online_Orders]), SUM(view_MasterKPIs[Total_Online_Customers]), 0)

Daily Sales Run Rate (DSRR):

- DIVIDE(SUM(view_MasterKPIs[Total_Online_Revenue]), 30, 0)

Average Products per Order (APPO):

- DIVIDE(SUM(view_MasterKPIs[Total_Online_Units_Ordered]), SUM(view_MasterKPIs[Total_Online_Orders]))

Average Order Value (AOV):

- DIVIDE(SUM(view_MasterKPIs[Total_Online_Revenue]), SUM(view_MasterKPIs[Total_Online_Orders]), 0)

Visualizations

Clustered Bar Chart#1:

- Top Countries by Revenue.
- Identifies highest-performing countries.
- Highlights key demographic market driving overall sales.

Clustered Bar Chart#2:

- Global Online Sales by Region.
- Shows which regions drive revenue.

- Displays top-performing regions.

Line Chart:
- Online Sales Trend Over Time.
- Shows seasonal patterns and growth trends
- Tracks monthly performance from 2010-204.

Column Chart:
- Total Units Sold by Category.
- Shows physical volume distribution across product categories.
- Reveals which categories have the highest unit movement.

Stacked Bar Chart:
- Profitability by Category.
- Shows which categories are most profitable.
- Displays Total Online Revenue vs Total Cost.
- Reveals profit margins by category.

Donut Chart:
- Product Category Mix.
- Shows revenue distribution across categories.
- Highlights specific product categorical dominance among other product categories.

Declaration of AI use

Stated below are the following AI assisted tools used in guiding the creation of this project:
- **Qwen AI:** Used for brainstorming, asking suggestions & guides in developing the dashboard.
- **Opencode (Desktop):** To analyze the database and produce README file schema report.
- **Claude:** Converted the README file into a proper PDF schema report file.

Profitability & Promotion

Dashboard Title

- ***AdventureWorks Promotions & Profitability Dashboard***

Business Goal

- Evaluate the effectiveness and profitability of the sales promotions done by comparing the sales, profit, profit margin, and discount rates to identify whether promotions should be continued or reassessed.

Critical Success Factor

- **Data Accuracy and Reliability:** Sales, profit, margin, and discount metrics reflect the underlying data to support decisions
- **Effectiveness of promotions:** Identify which promotions generate strong profitability while maintaining flexible discount levels.
- **Clarity:** Present performance through easy-to-interpret visuals to allow managers to identify strong and weak promotions.
- **Interactivity:** Allows users to select completed promotions and examine how each performed.
- **Actionability**: Insights from the dashboard to inform decisions for the betterment of the company.

Primary Users

- **Sales Managers:** Evaluate promotion performance and identify if promotions should be continued or adjusted.
- **Marketing Managers:** Evaluate whether the company's discounts are generating sufficient sales and profitability.
- **Business Executives:** Monitor overall promotional profitability and support resource allocation decisions.

Data Source

- Primary Data Source: SQL Server – AdventureWorksDW2025
- Tables used:
    - **Fact Tables:**
        - FactInternetSales
        - FactResellerSales
    - **Dimensions Tables:**
        - DimPromotion
        - DimProduct
        - DimProductCategory
        - DimProductSubcategory
    - **Supporting Table:**
        - Promotion Selector

Filters or Slicers

- **Promotion Selector:**
  - Allows users to select a promotion.
  - Updates the performance analysis based on the user's selected promotion.

- **Visual-Level Filtering:**
  - Updates the performance analysis based on the user's selected promotion.

Lagging KPIs

- **Total Sales** - This measures the total revenue generated from both online and reseller sales. Reflects the company's sales performance.

- **Total Profit** - Calculates total sales minus the product cost. Measures actual profit generated after accounting for product costs.

- **Overall Profit Margin** - Calculates total profit divided by total sales. Indicates how efficiently sales are converted into profit.

Leading KPIs

- **Average Discount Rate** - Weighted discount amount divided by total extended sales amount. This indicates the level of discounting applied to sales. Helps anticipate potential impact for future profitability.

KPI Computations

Lagging KPIs:
- **Total Sales** = Internet Sales + Reseller Sales
- **Total Profit** = Total Sales - Total Product Cost
- **Profit Margin%** = Total Profit / Total Sales * 100

Leading KPIs:
- **Average Discount Rate%** = Weighted Discount Amount / Total Extended Amount * 100

DAX Measures Reference

Total Sales All:
- SUM(FactInternetSales[SalesAmount]) + SUM(FactResellerSales[SalesAmount])

Total Profit All:
- (SUM(FactInternetSales[SalesAmount]) - SUM(FactInternetSales[TotalProductCost]))
  + (SUM(FactResellerSales[SalesAmount]) - SUM(FactResellerSales[TotalProductCost]))

Profit Margin All:
- DIVIDE([Total Profit All], [Total Sales All])

Total Discount All:
- SUM(FactInternetSales[DiscountAmount]) + SUM(FactResellerSales[DiscountAmount])

Discount Rate All:
- DIVIDE(SUMX(FactInternetSales, FactInternetSales[ExtendedAmount] * FactInternetSales[UnitPriceDiscountPct]) + SUMX(FactResellerSales, FactResellerSales[ExtendedAmount] * FactResellerSales[UnitPriceDiscountPct]), SUM(FactInternetSales[ExtendedAmount]) + SUM(FactResellerSales[ExtendedAmount]), 0)

Show Selected Promotion

- Returns 1 when no promotion is selected or when the current promotion matches the selected promotion; otherwise, returns 0.
- Filters the charts to the selected promotion

Bar Highlight Color:

- Dark blue if selected, otherwise light blue
- Highlights the selected promotion in the bar chart

Selected Promotion Bubble Size:

- 80 if selected, otherwise 20
- Makes the selected promotion larger in the scatter plot

Management Remarks:

- Promotion and reason
- No Data = No sales
- Underperforming = Negative Profit
- High Discount Risk = Positive Profit with a high discount rate
- Profitable = Positive profit with a reasonable discount

Visualizations

Clustered Bar Chart:

- Profit by Promotion
- Compares the profit generated by promotions present
- Highlights promotions with strong or weak profitability

Scatter Plot:

- Discount vs. Profitability
- Relationship between discount rate and profit margin
- Identifies promotions' relationship with discounts and profitability

Table:

- Promotion Performance Details
- Shows sales, profit, profit margin, and discount rate for each promotion presented.
- Users see a summary of the promotion and can view the selected promotion in greater detail.

Dynamic Remark:

- Management Remark
- Provides an automatic interpretation based on the selected promotion's performance.

Declaration of AI use

Stated below are the following AI assisted tools used in guiding the creation of this project:

- **OpenAI**: Used as a supplementary tool for clarifying Power BI and DAX concepts. Troubleshooting and refining

Customer Segmentation

Dashboard Title

- ***AdventureWorks Customer Segmentation Dashboard***

Business Goal

- Analyze customer demographics to identify who have a high purchasing power and who are frequently buying to provide promotions, discounts, and perks in maintaining customer retention and maximize the generated revenue.

Critical Success Factor

- **Customer Retention** - Encourages first-time buyers to keep on buying with no hesitation
- **Customer Lifetime Value Growth** - Increase customer volume to generate more revenue
- **Customer Demographics** - Develop customer profiles to curate what products they will be needed
- **Revenue Optimization** - Understanding different income ranges from customers to match their budget goods and services
- **Effective Marketing and Promotion** - Create marketing campaigns based on the customer demographics.

Primary Users

- **Marketing Committee** - Makes customer profiles and plans for strategies to maintain or improve customer retention rate.
- **Sales Committee** - Formulates sales policies and strategies to generate revenue in the company.
- **Customer Relations Officers** - Ensures that customers will be provided goods and services in a proper manner.

Data Source

- **Primary Data Source:** SQL Server - AdventureWorks Database 2025
- **Table Used:**
    - **Fact Tables:** FactInternetSales, FactSurveyResponse
    - **Dimensions Tables:**
        - DimCustomer
        - DimGeography

Filters or Slicers

- **Generation**
    - Greatest Generation
    - Silent Generation
    - Baby Boom Generation
    - Generation X
    - Millenial
- **Marital Status**
    - Single
    - Married
- **Country**
    - Australia
    - Canada
    - France
    - Germany

- United Kingdom
- United States

Lagging KPIs

- **Average Revenue per Customer** - Measures the purchase amount that an average customer will spend. The higher the average revenue is, the more effective their past sales were.

Leading KPIs

- **Customer Retention Rate** - Measures the purchasing power of customers who will buy again in some time. Shows how likely customers will stick to the e-commerce platform.
- **Survey Engagement Rate** - Measures the engagement of customers who have answered the survey. Calculates how trustful the administration or management are to the customers.

KPI Computations

Average Revenue per Customer

- **Formula:** $\frac{Total\ Revenue}{Number\ of\ Customers} \times 100\%$

Customer Retention Rate:

- **Formula:** $\frac{Number\ of\ Customers\ who\ ordered\ at\ least\ 2\ orders}{Number\ of\ Customers} \times 100\%$

Survey Engagement Rate:

- **Formula:** $\frac{Number\ of\ Surveys\ Answered}{Number\ of\ Customers} \times 100\%$

DAX Measures Reference

Average Revenue per Customer:

```
DIVIDE(SUM(FactInternetSales[SalesAmount]),
DISTINCTCOUNT(DimCustomer[CustomerKey]))
```

Customer Retention Rate:

```
VAR CustomerRetention =
    COUNTROWS(
        FILTER(
            VALUES(FactInternetSales[CustomerKey]),
            CALCULATE(DISTINCTCOUNT(FactInternetSales[SalesOrderNumber])) > 1
        )
    )
RETURN DIVIDE(CustomerRetention, DISTINCTCOUNT(FactInternetSales[CustomerKey]))
```

Survey Engagement Rate:

```
DIVIDE(
    DISTINCTCOUNT(FactSurveyResponse[CustomerKey]),
    DISTINCTCOUNT(DimCustomer[CustomerKey])
)
```

Generation:

```
SWITCH(
    TRUE(),
    DimCustomer[BirthDate].[Year] <= 1927, "Greatest Generation",
```

- DimCustomer[BirthDate].[Year] <= 1945, "Silent Generation",
- DimCustomer[BirthDate].[Year] <= 1964, "Baby Boom Generation",
- DimCustomer[BirthDate].[Year] <= 1980, "Generation X",
- "Millenial"
-)

Visualizations

Clustered bar chart
- Total Revenue by Occupation based from Gender
- Illustrates how much each gender has earned from their respective careers

Line Chart
- First-Time Customers Frequency over Time
- Displays the growth of AdventureWorks users in months or years.

Matrix
- Top 20 Customers who Frequently Purchase
- Shows a list of customers who regularly use the website
- These customers may be provided perks and discounts from the marketing team.

Clustered column chart
- Average Yearly Income by Number of Children
- Exhibits how yearly income depends on the number of children